Stat 565 Intermediate Report

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April 2026

Project Update and Scope

This report is the intermediate update for my replication project of *Topological Data Analysis of Financial Time Series: Landscapes of Crashes* by Marian Gidea and Yuri Katz (`https://arxiv.org/pdf/1703.04385`). The core goal remains the same as in my proposal: replicate the data pipeline and empirical behavior of topological early warning indicators before major US equity market crashes.

I have now moved from planning into implementation. The notebook is built and currently reproduces the main data engineering steps, the rolling TDA computations, and event-focused visual diagnostics around the dot-com and Lehman periods. At this stage, the code is working end-to-end and produces preliminary outputs that are consistent with what the paper claims qualitatively.

Data Description

The data source is Yahoo Finance (via Python package `yfinance`), matching the source used by the original paper. I downloaded daily adjusted close time series for:

- S&P 500 (`^GSPC`)
- NASDAQ Composite (`^IXIC`)
- DJIA (`^DJI`)
- Russell 2000 (`^RUT`)

I also downloaded VIX (`^VIX`) as a benchmark series for side-by-side comparison, because the original paper compares topological indicators with volatility behavior. The main sample is from 1987-12-23 through 2016-12-08 (end date handled in code as exclusive next day), which aligns with the date range described in the paper.

For each of the four equity index series, I compute daily log-returns using: $r_{i,j} = \ln\left(\frac{P_{i,j}}{P_{i-1,j}}\right)$, where $P_{i,j}$ is adjusted close for index j on day i. The resulting data matrix is a 4-dimensional multivariate return time series that becomes the input for all topological computations.

Preliminary Analysis

Before applying TDA, I checked the basic data behavior and alignment. The raw adjusted close levels and log-return series are available and have the expected properties (long-run trend in levels, noisy mean-reverting return fluctuations, and visible high-volatility regimes around known crisis periods).

Then I ran the rolling topological pipeline using windows $w = 50$ and $w = 100$, as in the paper's main setup. For each rolling window, I form a 4D point cloud, compute 1D persistent homology (H_1), build persistence landscape approximations, and track the L^1/L^2 norms over time. The generated norm series show spikes and clustering near stressed market periods, which is a good preliminary sign for replication quality.

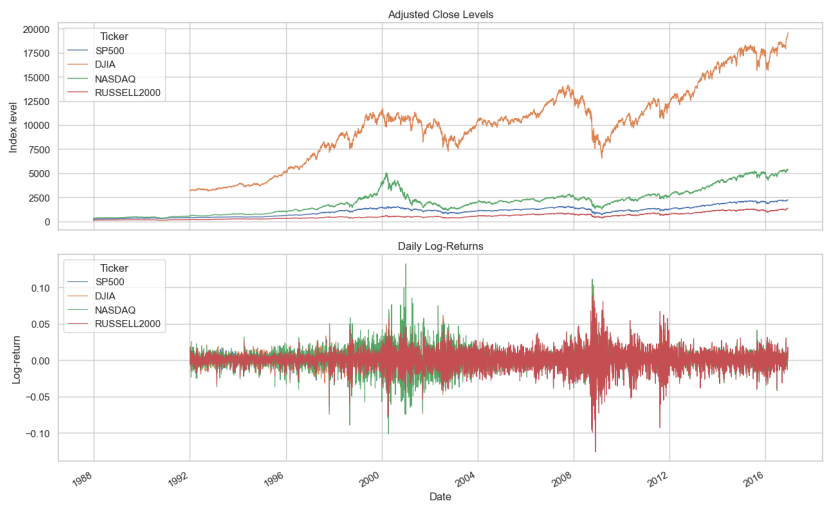

Figure 1: Preliminary data check from the notebook: adjusted close levels and daily log-returns for the four index series.

Methodologies Implemented

The methodology follows the original paper's framework with practical implementation decisions in Python:

1. **Rolling point-cloud construction.** For each date, I collect w consecutive 4D observations (four index log-returns), giving one point cloud in $\mathbb{R}^4$.

2. **Persistent homology computation.** I compute 1D persistence diagrams from each cloud using Vietoris–Rips filtration (via `ripser`, max dimension = 1).

3. **Persistence landscape norm approximation.** I numerically approximate persistence landscapes from birth/death pairs and compute L^1 and L^2 norms for each window.

4. **Early warning diagnostics.** On the resulting norm time series, I compute rolling variance and low-frequency spectral power, and apply Mann–Kendall/Kendall-τ style trend checks on pre-crash windows.

I am keeping this implementation close to the paper while still using modern Python tools rather than the original R package stack. This should make the replication both transparent and easier to extend later.

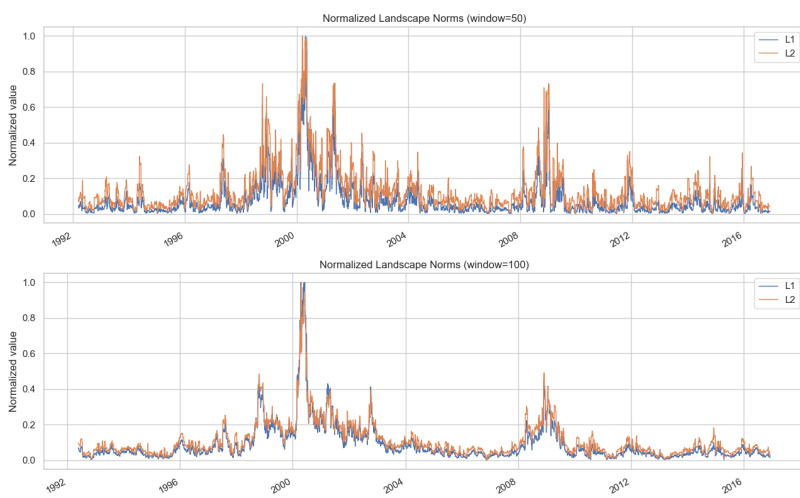

Figure 2: Normalized persistence landscape norm series (L^1, L^2) from rolling windows (preliminary replication output).

Expected Results

Based on current outputs and on the target paper, I expect the final report to show the following:

- The time series of persistence landscape norms should increase sharply near major stress periods, with noticeable fore-shocks before the main crash dates.
- Rolling variance and low-frequency spectral density of the norm series should exhibit positive trends in the 250 trading days before 03/10/2000 and 09/15/2008.
- The topological indicators should provide complementary information to VIX (sometimes sharper, sometimes similar), rather than simply duplicating VIX behavior.
- Window size changes ($w = 50$ to $w = 100$) should preserve the broad signal structure but potentially smooth peaks and shift maxima slightly, as discussed in the paper.

In plain terms, I expect to recover the main qualitative claim: TDA-based persistence landscape norms can act as a useful early warning signal class for imminent market crashes. Quantitatively exact figure matching may differ a bit because of implementation details (library differences, numerical approximations, and data vendor updates), but the directional behavior should remain consistent.

Next Steps

- Finalize figure-by-figure alignment with the paper's event windows and normalization conventions.
- Add robustness checks for parameter choices (grid size, filtration effects, and potential scaling sensitivity).
- Cleanly document reproducibility details (package versions, run order, runtime notes).
- Prepare final write-up linking each reproduced result to the corresponding original figure/claim.